

Project Documentation: MaintainIQ

1. Project Overview

MaintainIQ is an AI-powered smart diagnosis SaaS application designed to streamline clinic management and optimize patient record tracking. It integrates intelligent diagnostic tools to assist medical practitioners in real-time.

2. Live Deployment Links

Frontend (Live): [frontend](#)

Backend (API): <https://maintainiq-hackathon.onrender.com>

GitHub Repository: <https://github.com/Abdullah-ShahAlam-Dev/maintainiq-hackathon>

3. Key Technical Features

- Role-Based Authentication: Secure JWT-based access for admins, doctors, and patients.
- Smart Diagnosis Module: AI-driven insights to assist in patient record analysis.
- Patient Timeline Tracking: Automated history management for clinical records.
- Document Generation: Automated generation of prescription documents.

4. Tech Stack

- Frontend: React, Tailwind CSS
- Backend: Node.js, Express.js
- Database: MongoDB
- Deployment: Vercel (Frontend), Render (Backend)

5. Implementation Details

Authentication: Implemented JWT authentication and password hashing for high-security standards.

Data Synchronization: Managed hybrid data flow between localized storage and the centralized MongoDB database.

API Integration: Successfully established cross-origin communication between the frontend client and the Render-hosted backend server.